Dry Mass Measurement Softwaretool - Manual

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How to Navigate and Utilize This Manual

This manual should contain a detailed description of how to work with the respective software tool. It is structured so that the titles of the sections provide an initial indication of how the steps of the software tool are to be carried out. This means that the user already understands the most important steps when reading the table of contents.
The manual starts with an introduction followed by a general description of the structure of the software tool. This is followed by a more detailed explanation of all functions and requirements.
Information about the installation is here not provided; this can be found in the README.md on the github repository.

Contents

1 Introduction

This software provides an easy and straightforward method to determine the dry mass of a single cell. The Transport of Intensity Equation (TIE) is used to determine this property. This method requires only a stack of images taken with a widefield microscope.
Two different solutions are implemented to solve the TIE: Fast Fourier Transform (FFT) and Total Variation (TV) Regularization. In addition, contour detection is implemented to determine the area of a cell.
In summary, this software tool provides methods to determine the cell dry mass and the cell area.

2 An Overview of Software Tool Operations

The software tool is so designed that it can only go further in the calculation if all the steps are done in the right sequence. That means all of the buttons that can not be used are disabled. If you want to make a step where the button is disabled, it means that a step in the sequence is missing. The steps have to be done as described in this manual, which can already be seen in the table of contents.
The software consists of four parts: uploading a file and setting the parameters, selecting the cell as precisely as possible, calculating the optical path delay (OPD) and adjusting the contour for calculating the area.
The software can adjust all generated plots and images with the corresponding toolbar. Detailed documentation can be found on the official page of the matplotlib library.

3 Uploading Files

It is possible to upload two file types: .lif and .tif. To upload these files, use the *'Upload file'* button in the toolbar - a dialog will appear. At the bottom right corner, you have to choose a filter for the file type you want to upload 1. When the upload is successful, you will be redirected to the settings.

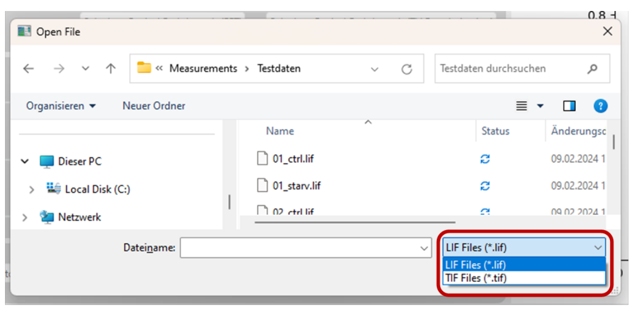

Figure 1: Uppload Process is shown; in the bottom right corner is a filter to choose the file type.

4 Settings

The settings window can be opened by clicking on the '*Settings*' button in the toolbar. The settings window also opens when a new file is successfully uploaded.
A background image is required for the calculation to remove the characteristics of the light later in the calculation, which is important to reduce artifacts in the phase reconstruction.
There are small differences in the upload window when choosing a lif file or a tif file, this is explained in section 4.1 and section 4.2. Firstly the settings, which are the same for both file types, are explained.

The parameters for cell dry mass calculation are are the following (all parameters have to be filled out):

- **Magnification** is the used microscope magnification used during the measurement.

- **Pixel size** refers to the physical size of each individual pixel in the camera sensor. (in nanometer)
- **Axial Step** is the distance moved by the microscope stage or the focal plane along the z-axis between each image in the z-stack. (in nanometer)
- **Constant** α is a tabulated constant for different proteins that is used in the calculation of their mass based on changes in the refractive index. (in ml/g)
- **Regularisation Constant** λ is the penalty factor in the TV-regularisation and controls the influence of the penalty function on the phase reconstruction (Too high λ values result in falsification regarding the cell mass. Therefore, it should be chosen as smallest as possible). The input is by a factor of 10^6 bigger than the value used for calculations.
- **Number of Iterations for TV-Norm** is the number used for the TV-regularisation method. A low number of iterations results in a bad phase reconstruction. A high number of iterations does not change the mass but results in a time-consuming calculation.

After setting all the parameters and choosing the files like explained in section 4.1 and section **??** the button '*Continue*' can be clicked for the next steps in the calculation. The upload process stops by hitting the quit button at the top right (x) and does not change the parameters. If no file was uploaded, the setting window closes, and no further calculations can be done.

4.1 Settings-.lif-file

When using .lif files, the background and sample images must be saved in the same .lif file. It is also possible that the file contains multiple measurements. It can be selected in the drop-down menu by clicking on the 'v' sign next to the file name that will be used for the calculation, shown in figure 2 and figure 3. The sample and background image must be selected in relation to their field names. Otherwise a negative mass will be calculated.
The desired file can be selected by mouse click.

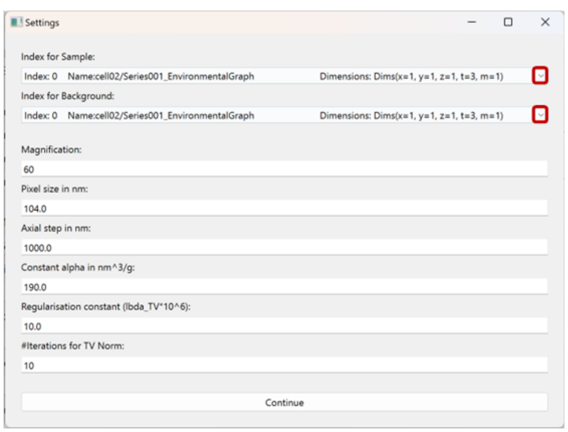

Figure 2: The button for opening the drop down menu is marked in red

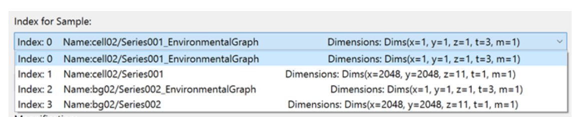

Figure 3: The drop down menu is shown, where the files for the calculation have to be chosen in relation to their field name. (Be careful to ensure that you do not chose the metadata)

4.2 Settings-.tif-file

If you use a .tif file, the background and sample images must be two separate files. By selecting the sample file during the upload 3, the background image needs to be uploaded in this step additionally. However, the sample file can still be changed here if something went wrong during the upload process. To upload a sample or background file, the '*upload sample file*' or '*upload background image*' button must be selected 4. An upload dialog will appear. Both files must be .tif files.

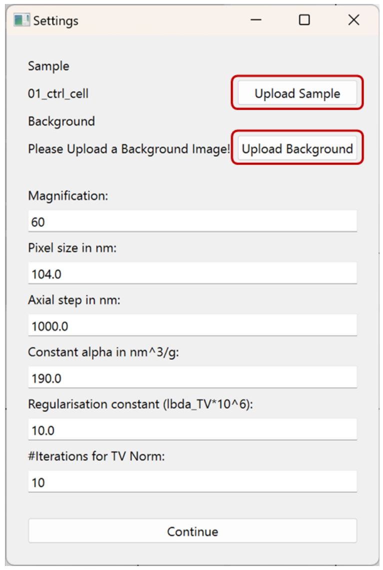

Figure 4: Settingwdinow, if a .tif-file is uploaded with buttons, marked in red, to upload files additionally.

5 Choosing the focused image

Once the files have been successfully uploaded, the index of the focused image is calculated. This index can be seen in the mainwindow **??**. You can check if the calculated index is correct by clicking on the '*Select focused image*' button in the toolbar. It is possible to go through the stack to select the correct one. The calculated index of the focused image is shown in the top center. By pressing the button '*Select this image*', figure 5, this image will be taken as the new focused image and the window will be closed.

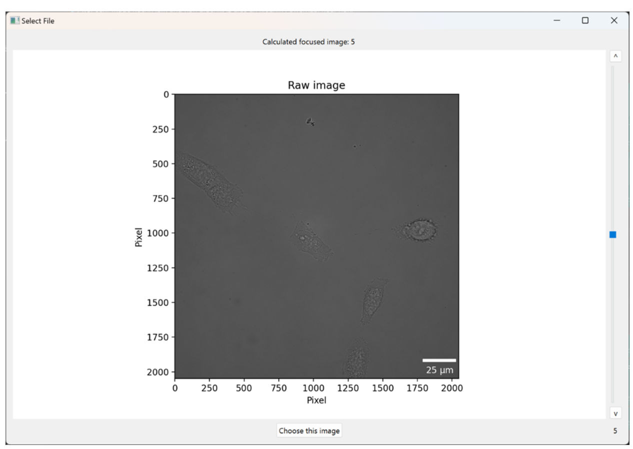

Figure 5: This is the window to select the focused image. The calculated index can be seen at the top center. At the right side with the slider you can go through the stack. To choose an image, you can click on the '*Choos this image*'.

6 Cell selection

For further calculations, selecting the cell as closely as possible is important. This step is very sensitive to calculations and has to be done carefully.
There are three different views, which can be shown by clicking on the buttons above the image:

- **Background Image** shows the background.
- **Image** shows the sample image.
- **Stack Image** shows the sample image divided through the background image.

For all views, an out-of-focus image is used for better visibility of the cells. The stack view is useful for checking if no cells from the background and the sample overlap.
Click on the image to display a red square, shown in figure 6. This square must be used to select the cell as precisely as possible. There must be no other cells or artifacts at the edges of the red square. Otherwise, the phase reconstruction will not work. Ensure there is also no cell in the background within the selected part. The stack view is useful for this.
To delete the selected part or draw a new one, click out of the red box and draw a new area.
If you change the view, the selected part is deleted.

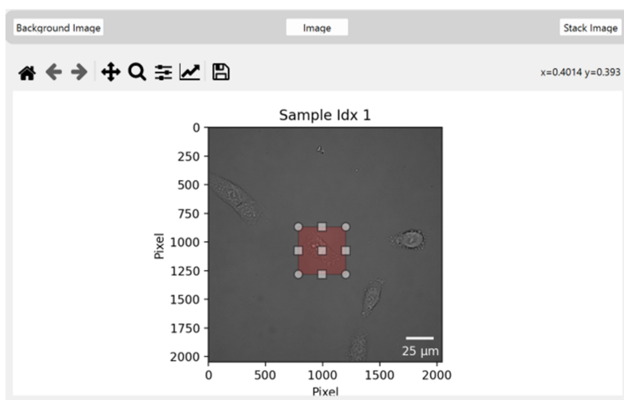

Figure 6: The process of selecting the cell is demonstrated. The red square has to be adjusted as precisely as possible with no artifacts/cells in the boundaries.

7 Dry Mass Calculation

The button and thus the dry mass calculation can only be carried out if an area is correctly defined, see section 6.
Two options are available for calculating the optical path delay (opd). The Fast Fourier Transform (FFT) 7.1 and the Total Variation (TV) Regularization 7.2. For both calculation methods, it is important that there are no cells or other artifacts (dust, noise, air bubbles, etc.) at the boundaries of the selected area.

7.1 FFT

It is important to enter an axis distance for the calculation. This parameter is used to define the axial distance for the derivative calculation. The axial distance can only be selected so that the stack is not exceeded during addition with the focused image and cannot be less than 0 during subtraction. A larger axial distance leads to a blurred effect in the reconstruction. In order to obtain the best possible solution, a small axial distance

should be tried. If the image is noisy or the reconstruction does not work properly, the axial separation can be set to larger distances, which results in less influence of the noise on the reconstruction.
The axial distance can be entered as an index and refers to the distance between the indices in the stack. After setting the axial distance, the optical path length can be calculated by clicking the '*Calculate Optical Path Length (FFT)**' button. If the axial distance is within the range, the optical path delay is calculated and displayed in the lower left-hand graphic.

Figure 7: This screenshot shows the entry field for the axial separation and plot for the optical path delay marked in red.

7.2 TV

For the calculation with TV regularization, it is important to set the parameter in the Settings Window 4 (the entry for the axis separation has no influence on this method).
This method is useful if the reconstruction with the FFT method 7.1 does not work. It is better suited to processing noisy images.
The penalty factor is crucial here. Penalty factors that are too large lead to a flattening effect, which results in a falsification of the dry mass. This factor is a compromise between flattening the cell mass and poor reconstruction. The best choice must be tried out manually. Iterations can also be set in the Settings Window 4, high iterations take a lot of time to reconstruct, a good estimate here is about 50 iterations, but this can vary for different images. In general, with this method, it is important to minimize the penalty factor to avoid falsifications in the dry mass. In addition, too few iterations lead to a poor reconstruction, while too many iterations do not influence the result.
After setting the parameters, the reconstruction can be executed by clicking the '*Calculate Optical Path Length (TV Regularization)*' button.

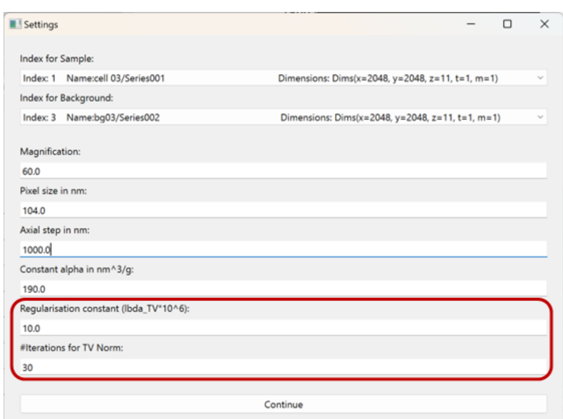

Figure 8: The cells to set the parameters for the TV regularisation in the Settings Window are shown here, marked in red.

Figure 9: The button to make the reconstruction with the TV regularisation is marked here.

8 Contour Detection

The opd must be calculated for contour detection (that means the bottom left plot can not be empty). The options for selecting a contour are listed here as subsections. The mass of the contour displayed in the plot is calculated by pressing the '*Calculate contour mass*' button; this process is the same for all contour recognition options.

8.1 Generate Contour

The contour detection has already taken place and is shown as yellow lines in the graphic. At the beginning, all possible contours are displayed.
Now it is important to set the parameters for contour detection so that one of these contours fits perfectly into the cell. There are two parameters for this. The '*Threshold*' can be seen as a parameter for controlling the sensitivity of contour detection. If this parameter is increased, the contour is only drawn if large intensity gaps exist. The lower the threshold value is selected, the more sensitive the contour detection is for small intensity gaps, figure 10.

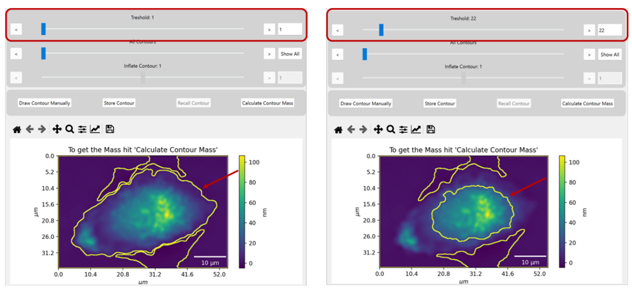

Figure 10: Contour detection is shown with two different parameters for the threshold.

Once the correct threshold value has been found, the contour can be selected using the '*Contours*' slider. If you move the slider all the way to the left, all contours are displayed. The contours are displayed individually if you move the slider to the right. A contour must be selected for the further calculation steps, figure 11.

If the contour found does not cover the entire cell, the contour can be inflated or deflated, shown in figure 12.

Once you have found the correct contour, you can calculate the mass within the contour by clicking on the '*Calculate Contour Mass*' button.

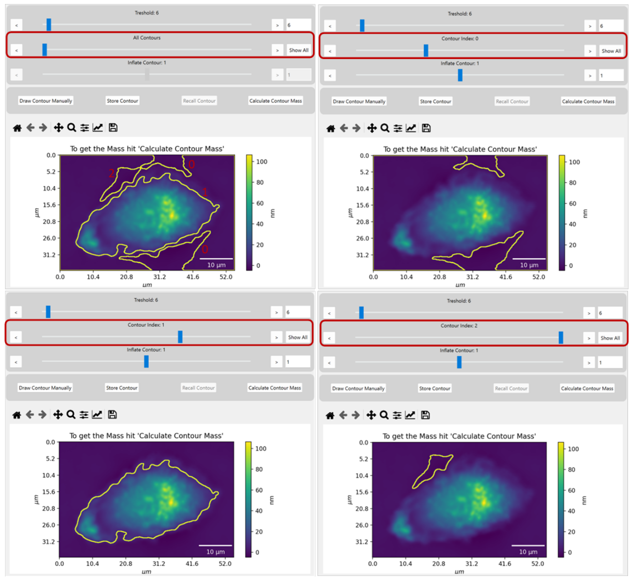

Figure 11: As an example, it is shown how to skip through the contours individually.

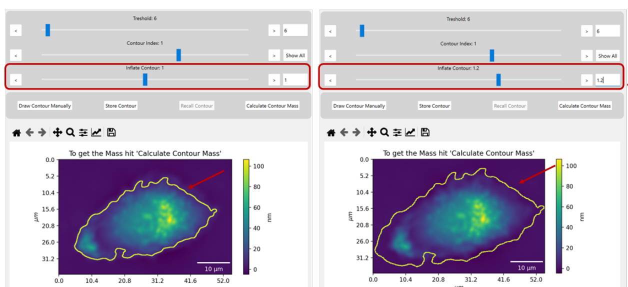

Figure 12: The inflate process can be seen in this image.

8.2 Draw Contour Manually

If no contour can be found when setting the parameter correctly, it is possible to draw the contour manually by selecting the '*Draw contour manually*' button. The created contours are hidden and by clicking with the right mouse button on the image, points can be drawn to set the contour. To delete this contour, click on the image again and start drawing. When you release the mouse button, the shape is drawn 13. When satisfied with your drawn contour, click the '*Calculate contour mass*' button. This step calculates the mass of the contour. Click the '*Generate contour*' button to exit manual contour drawing mode.

8.3 Store Contour

Contours can also be saved for the time during which no new file is uploaded. To save, click on the '*Save contour*' button. To retrieve the contour, press '*Retrieve contour*'. If the retrieved contour is known, the button name changes to '*Hide saved contour*' and is highlighted in blue. It can only be saved a single contour. To save a new contour, hit the '*Save Contour*' button again.

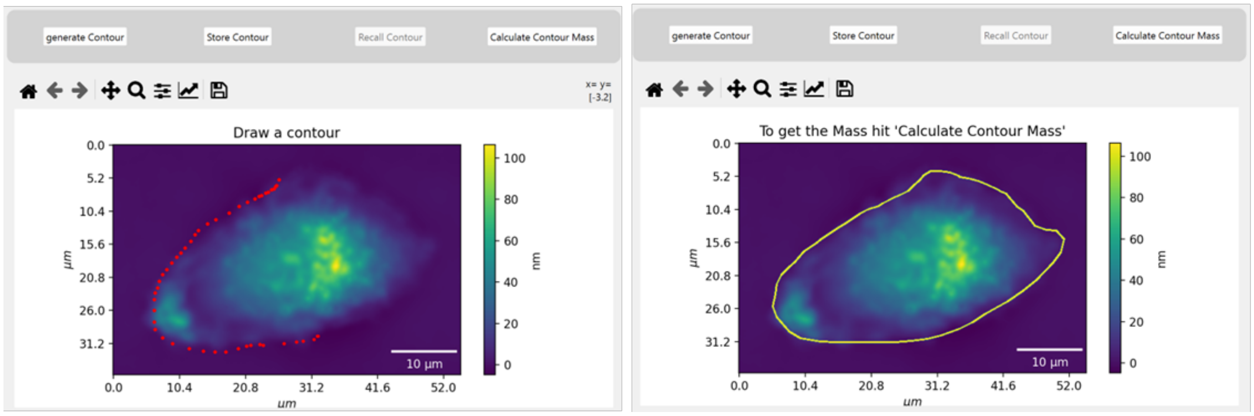

Figure 13: At the left, the process during the drawing mode can be seen; the red dots set the shape of the contour. By releasing the button, the contour gets drawn. The right image shows the contour when you release the right mouse button

Figure 14: The button change of the stored contour can be seen here. When the button is blue, the stored contour is displayed.

9 Saving

After calculating the mass, it is possible to save the calculated mass and the corresponding parameters. Clicking on the '*Save*' button in the toolbar opens a dialog for saving the files. A .csv file is saved with the important parameters for reconstructing the calculation; in addition, all three plots are saved in the same directory.

10 Evaluation

In the toolbar, there is a button '*Evaluation**', this is a small tool for plotting calculated data points. By clicking on the button, a window appears, with the button '*Upload file*' several files exported from the software tool can be selected, the areal mass density (mass/cell area) is then plotted. The average areal mass density is calculated and displayed next to the upload button.

11 Statistics

The statistics function is a tool for calculating the cell mass with various parameters; therefore, all calculation steps and contour detection steps previously mentioned have to be done before. The image size can be changed (in +x, -x, +y, -y direction). In addition, the axial distance for the FFT calculation and the parameters for the TV regularisation can be set. Separated with whitespace, it is possible to define multiple parameters in one step. To execute the calculation, the checkbox next to the input fields has to be checked. A separate directory is created in the path for each "statistic" performed. This directory has the same name as the file uploaded in the application. A metadata file is saved in this directory, which contains the 'names' of the calculated cells, the opd plot, and the .csv file, which are also saved in directories. This allows easy access for further analysis with other tools, making it easier to upload there.